BIASLENS — COLLABORATOR TESTING & HANDOFF GUIDE

Automated Bias Detection • Fairness Auditing • MS Word-Style Resume Editor • Dual-Engine Architecture

Target Repo: github.com/Sanchit-Agarwal-27/ResumeBiasDetector | Branch: main (Synced)

1. OVERVIEW & REPOSITORY STATUS

BiasLens is an AI-powered resume bias detector, fairness auditing engine, and interactive MS Word-style document editor. All latest updates (dynamic bundle splitting, GDPR privacy compliance, universal writing quality scanner, sticky review UI, and automated evaluation suites) have been verified and pushed to GitHub main.

2. CLONE OR PULL LATEST CODE

If you already have the repository cloned, pull the latest commits. If cloning fresh, use the standard clone command:

TERMINAL — GIT SETUP

```
# Option A: Pull latest code if already cloned
git fetch origin
git pull origin main

# Option B: Clone fresh repository
git clone https://github.com/Sanchit-Agarwal-27/ResumeBiasDetector.git
cd ResumeBiasDetector
```

3. FRONTEND SETUP & RUNNING

Prerequisites: Node.js (v18.0+) and npm. BiasLens functions 100% out of the box with zero required environment configuration. If Supabase keys are not provided, it automatically engages an offline local guest mode.

TERMINAL — FRONTEND COMMANDS

```
# 1. Install frontend dependencies
npm install

# 2. Run automated validation suite (57/57 tests)
npm test

# 3. Start local development server
npm run dev
```

Once started, open your browser at http://localhost:3000 (or the local URL printed by Vite in your console).

4. OPTIONAL: PYTHON FASTAPI AI MICROSERVICE

BiasLens uses a Dual-Engine Architecture. While the browser engine runs 100% client-side, an optional high-throughput Python FastAPI microservice is available in backend/ for deep NLP & ATS simulations:

TERMINAL — BACKEND SERVICE

```
cd backend
# Create virtual environment (Windows: python -m venv .venv | Mac/Linux: python3 -m venv .venv)
python -m venv .venv
# Activate environment: Windows -> .venv\Scripts\activate | Mac/Linux -> source .venv/bin/activate
.venv\Scripts\activate
pip install -r requirements.txt
pytest
uvicorn app.main:app --reload --port 8000
```

To connect: In the web UI, open Settings (gear icon in header) and switch Analysis Engine to 'Python FastAPI Microservice'.

5. INTERACTIVE FEATURE TESTING CHECKLIST

- **Interactive Word-Style Document Editor**
  Test rich text formatting, font families, font sizes, headings, text alignment, and bullet lists. Import existing PDF/DOCX resumes or click 'Load Sample Resume' in the toolbar to populate sample content.
- **Real-Time Context-Aware Bias Detection**
  Type or inspect highlighted bias tokens (e.g., gender-coded terminology, ageist phrases, socio-economic signals). Click any inline highlight or review item in the sidebar to review explainable AI reasoning and apply neutral rewrites with one click.
- **Writing Quality & Resume Hygiene Scanner**
  Switch to the 'Quality' tab in the Review Queue. Test automatic detection of passive voice, weak action verbs, unquantified claims, and corporate buzzwords.
- **Dynamic Google XYZ Sentence Rewriter**
  Click 'Rewrite with Google XYZ Formula' on resume bullet points to convert qualitative duties into quantified impact statements: Accomplished [X] as measured by [Y], by doing [Z].
- **Counterfactual Fairness Audit (EEOC 80% Rule)**
  Click 'Fairness Audit' in the navigation bar to trigger 10 demographic counterfactual perturbations (gender, race, age, non-traditional backgrounds) and check Disparate Impact compliance.
- **Empirical Model Benchmarks Modal**
  Click 'Benchmarks' in the navigation bar to run and inspect live confusion matrices, precision, recall, and accuracy metrics on the 30-sample gold dataset.
- **Vector PDF & Clean DOCX Export**
  Click the 'Export' dropdown in the editor toolbar to download the cleaned, de-biased resume as a publication-ready Vector PDF or editable DOCX.